LAB ASSIGNMENT 1
Operating Systems (UCS-303)

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Section: MCA-2

1. Introduction to the operating system with key functions.

Answer:

An Operating System (OS) is system software that acts as an interface between the user and computer hardware. It manages hardware resources and provides an environment for running application programs.

Key Functions:

- Process Management: Manages the creation, execution, scheduling, and termination of processes.
- Memory Management: Allocates and deallocates memory for running programs.
- File Management: Organizes, stores, retrieves, and protects files and directories.
- Device Management: Controls input and output devices using device drivers.
- Security and Protection: Protects system resources and data from unauthorized access.
- User Interface: Provides interfaces such as CLI and GUI for users to interact with the system.

2. Introduction to the Unix/Linux (Architecture).

Answer:

Unix/Linux follows a layered architecture consisting of the following main components:

- Hardware: Includes CPU, memory, storage devices, and input/output devices.
- Kernel: The core of the operating system that manages hardware, memory, processes, devices, and system calls.
- Shell: Provides an interface between the user and the operating system by interpreting commands.
- Applications and Utilities: Includes user applications, system utilities, editors, compilers, and other programs.

Basic structure:

Hardware → Kernel → Shell → Applications

3. Concept of Shell.

Answer:

A **Shell** is a command-line interpreter that provides an interface between the user and the operating system. It accepts commands from the user, interprets them, and executes the required operations.

Main functions of a Shell:

- Executes system commands.
- Manages environment variables.
- Provides command history and other interactive features.
- Supports shell scripting for automation.

4. Types of Shell.

Answer:

Common Unix/Linux shells include:

- Bash (Bourne Again Shell): A widely used shell and the default shell in many Linux distributions.
- sh (Bourne Shell): An early Unix shell used for command execution and scripting.
- Zsh (Z Shell): An enhanced shell with advanced features such as auto-completion and customization.
- Csh/Tcsh (C Shell): A shell with syntax influenced by the C programming language.
- Ksh (Korn Shell): A shell that combines features of the Bourne shell with additional scripting capabilities.

5. Command structure.

Answer:

Linux commands follow a consistent and standardized syntax structure:
Syntax: command [options] [arguments]
Components:
Where:

- Command: Specifies the operation to be performed.
- Options: Modify the behavior of the command.
- Arguments**:** Specify the files, directories, or other inputs on which the command operates.

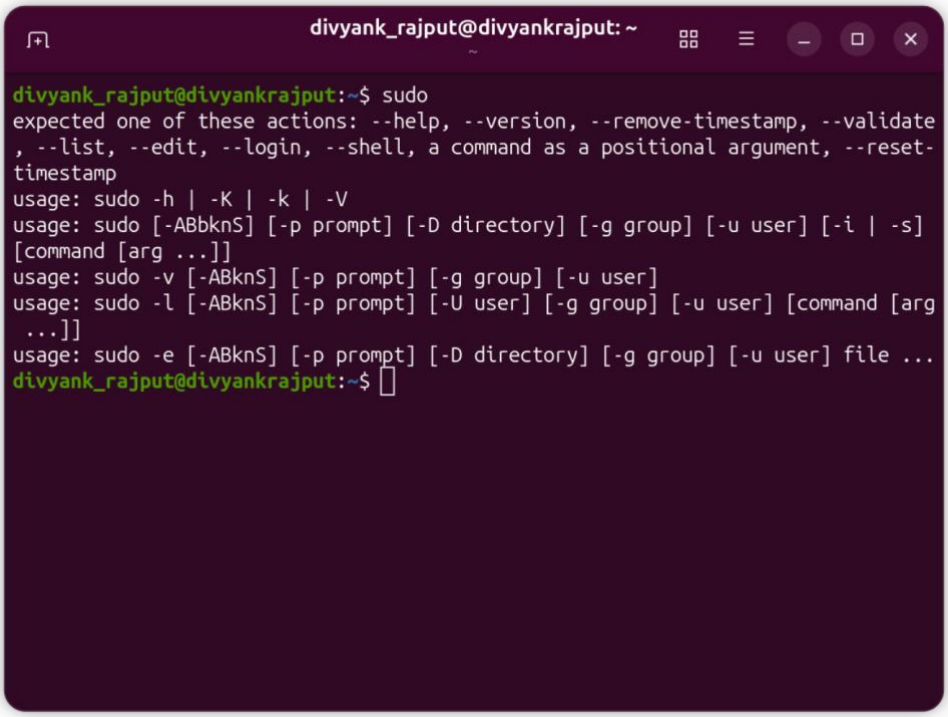
Example Breakdown:
ls -la /var/log

- ls: The command to list directory contents.
- -la: Options (-l for long detailed format, -a to show hidden files).
- /var/log: Argument specifying the target directory path.

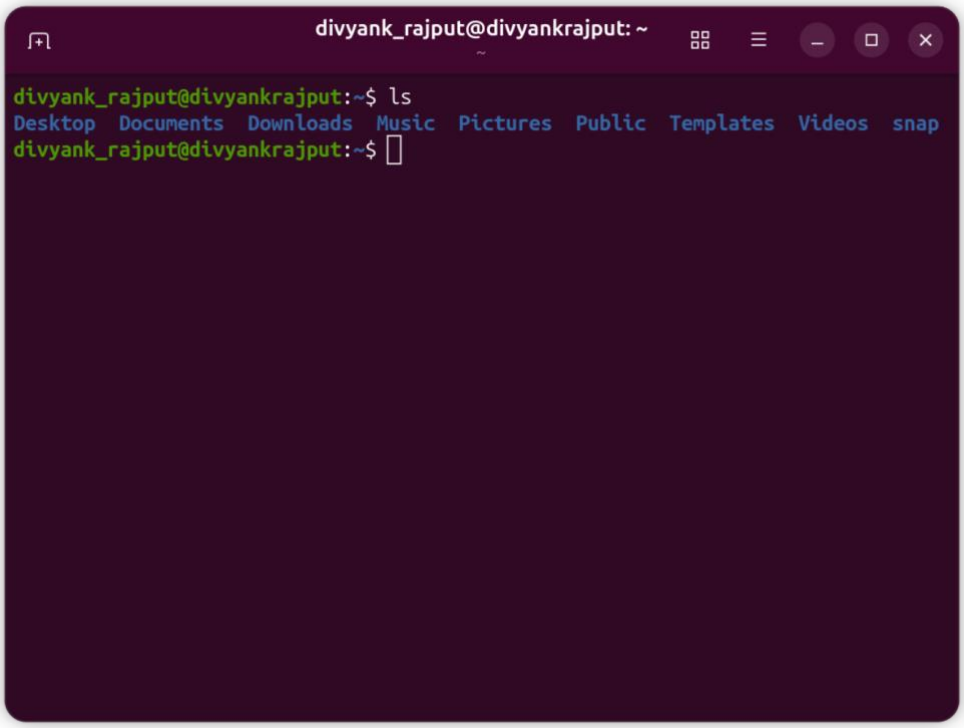
6. Introduction of basic linux commands (sudo, ls, pwd, mkdir, rmdir, rm, cd, cp, wc, mv, cmp, passwd, who, uname)

Answer

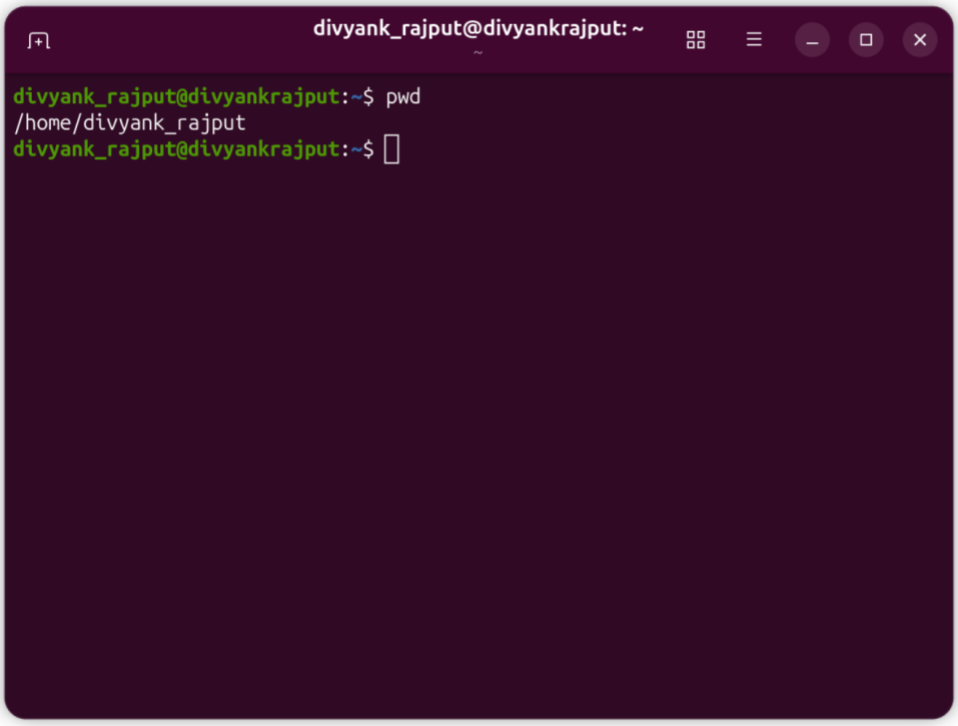
- sudo - Used to execute commands with superuser or administrative privileges.



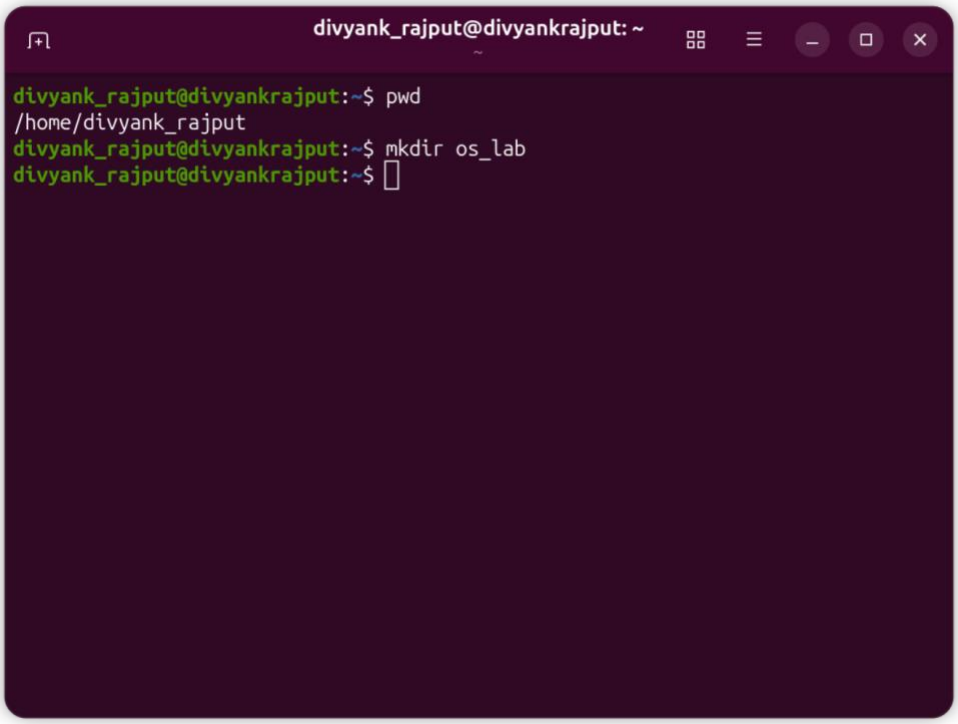
- ls - Used to list files and directories in the current directory.



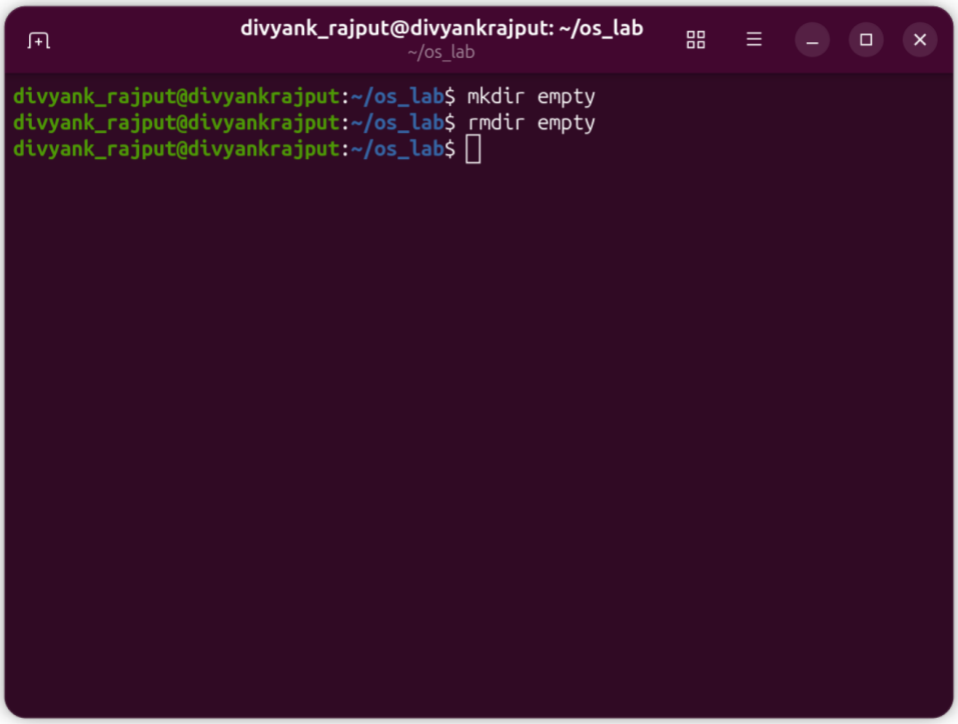
- pwd - Used to display the path of the current working directory.



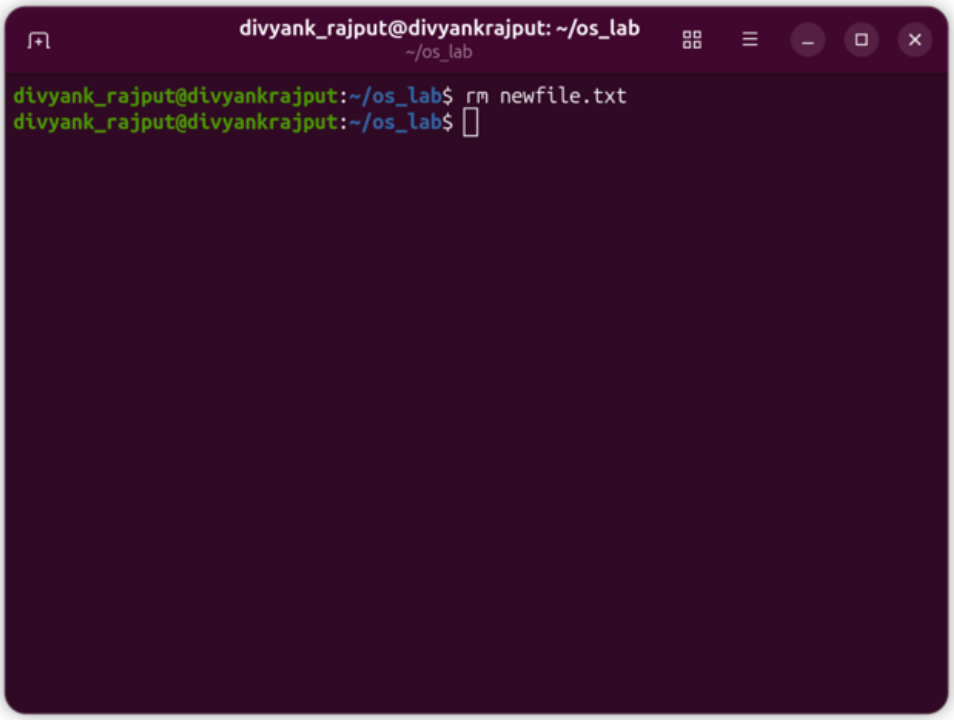
- mkdir - Used to create a new directory.



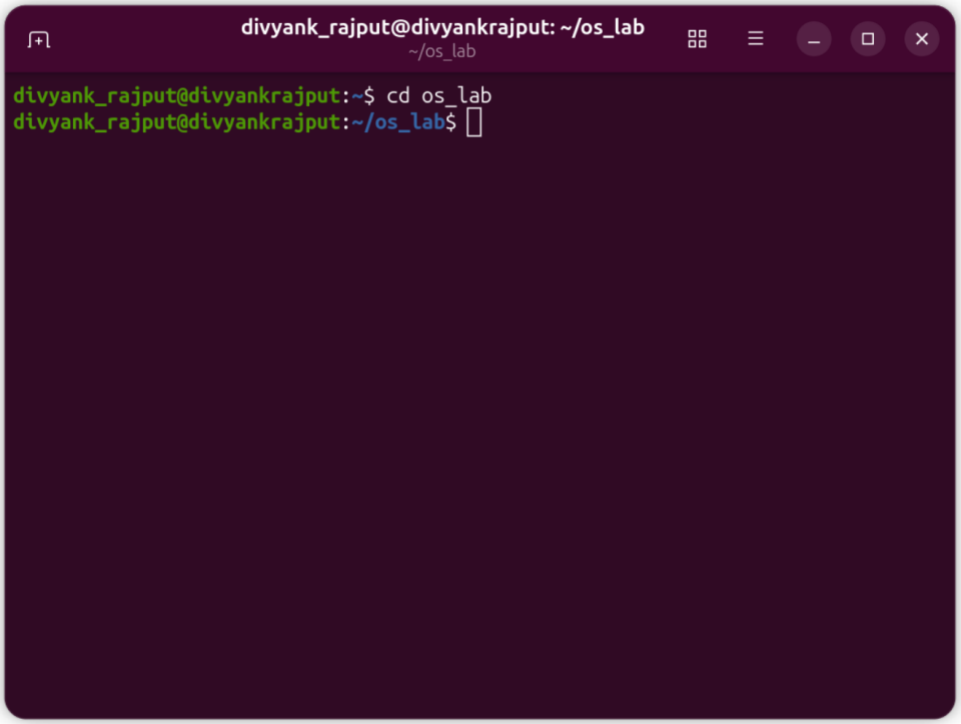
- rmdir - Used to remove an empty directory.



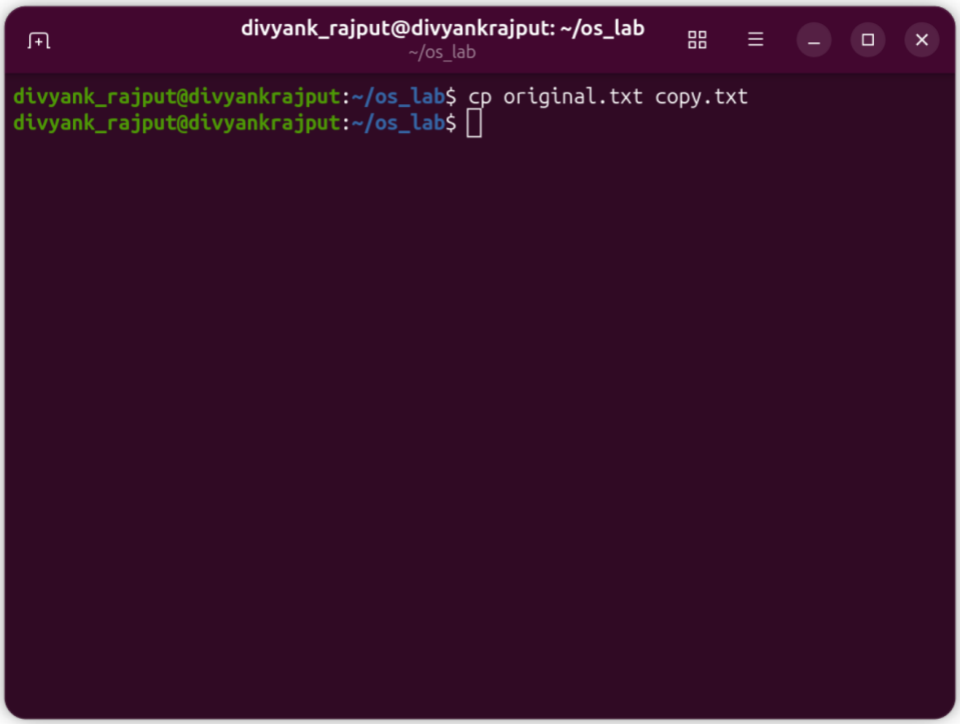
- rm - Used to remove files or directories.



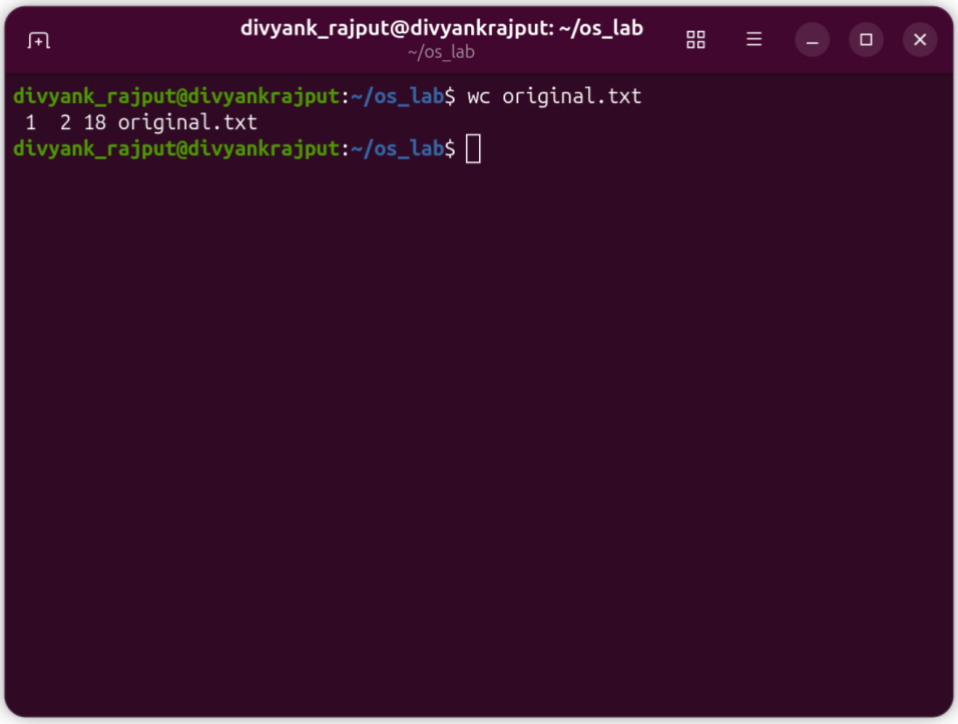
- cd - Used to change the current working directory



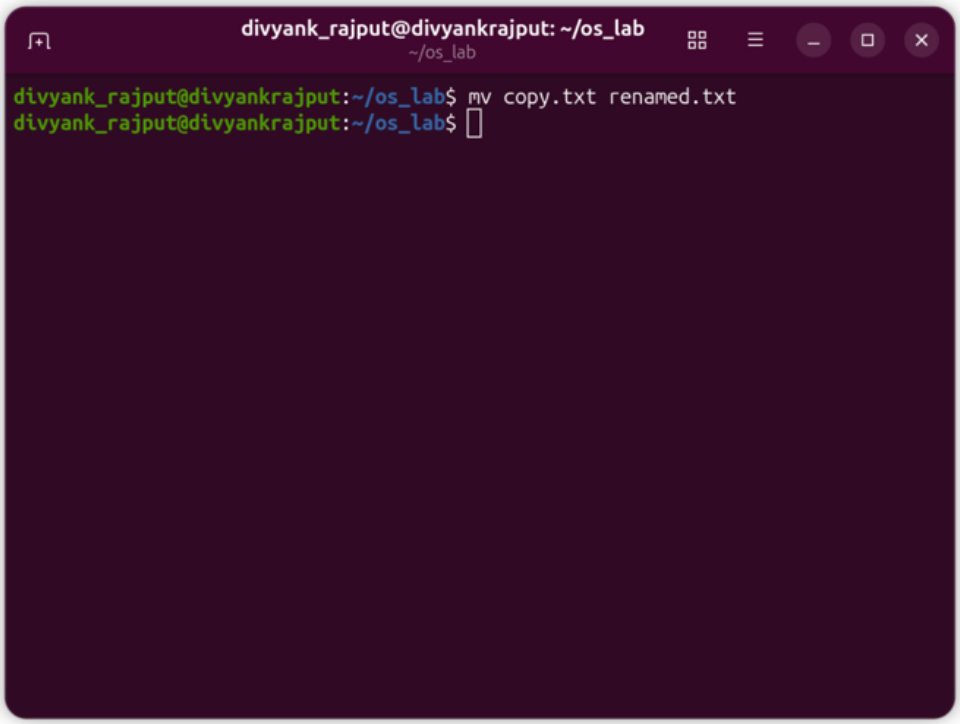
- cp - Used to copy files or directories from one location to another.



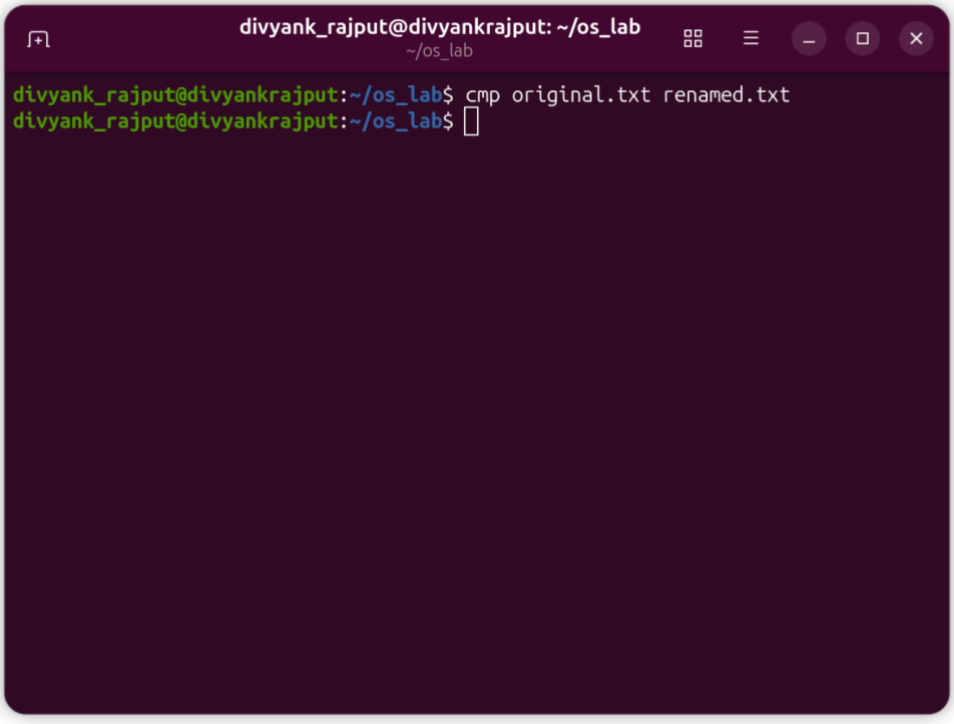
- wc - Used to count lines, words, and bytes in a file.



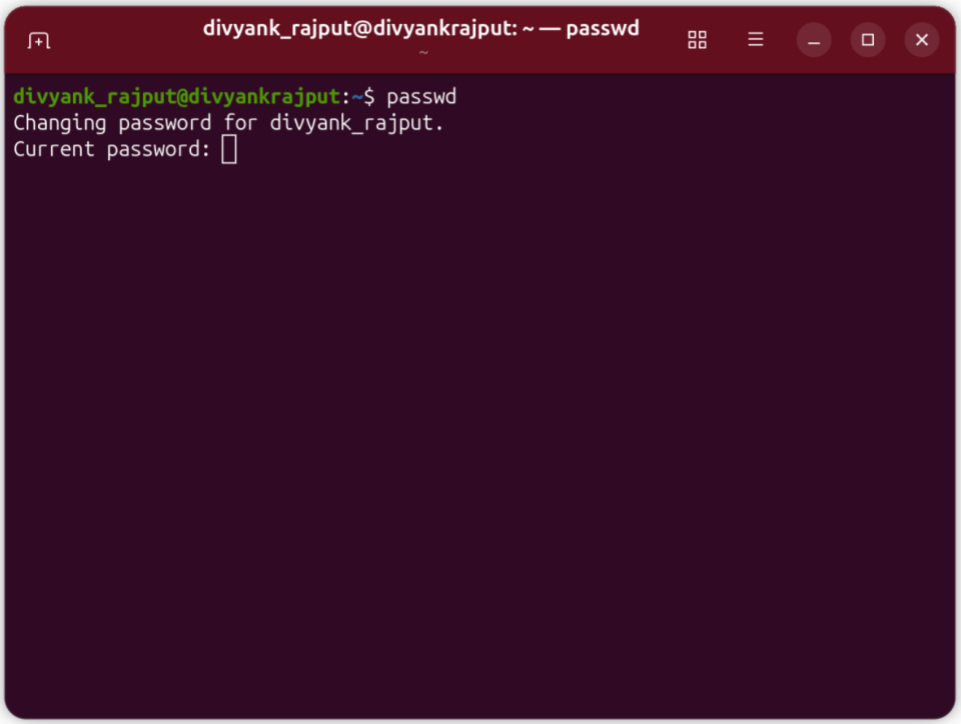
- mv - Used to move or rename files and directories.



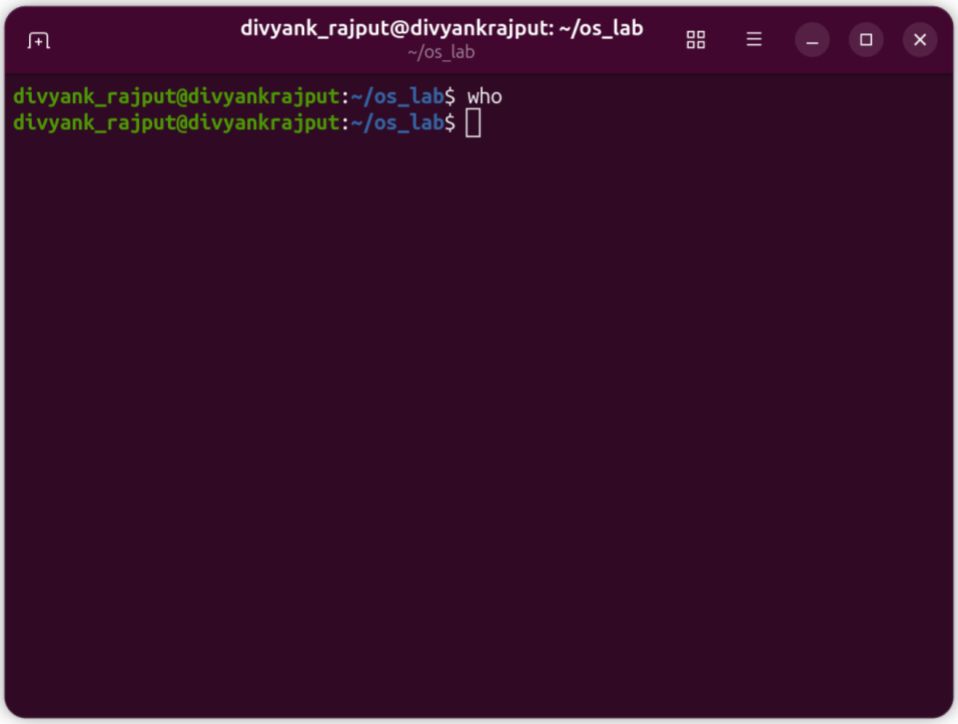
- cmp - Used to compare two files and identify differences between them.



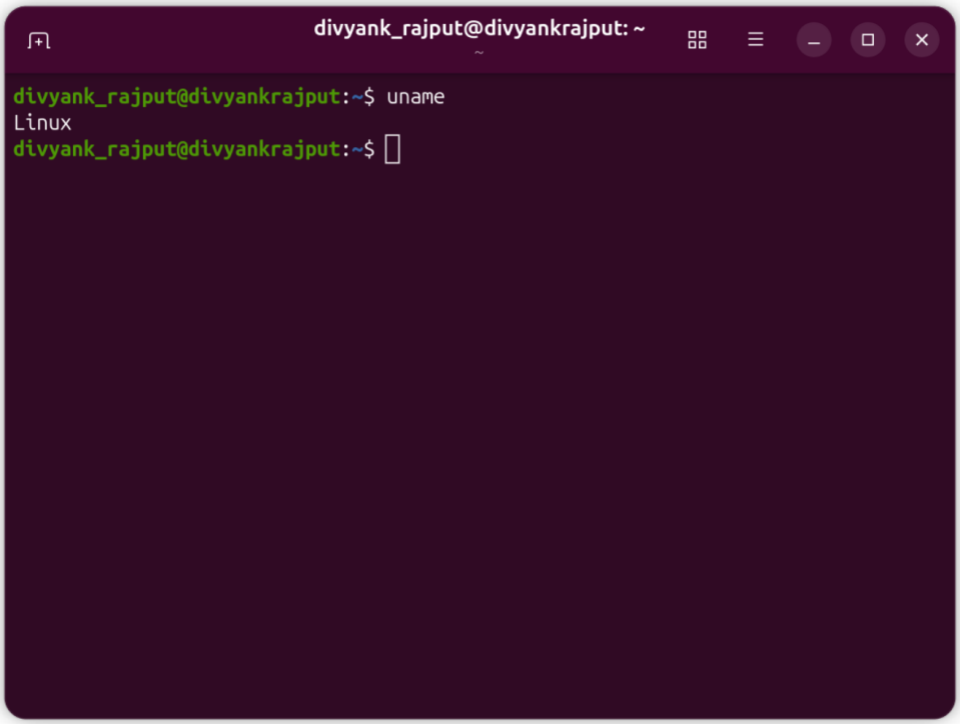
- passwd - Used to change the password of a user account.



- who - Used to display information about users currently logged into the system.



- uname - Used to display basic information about the operating system and kernel.

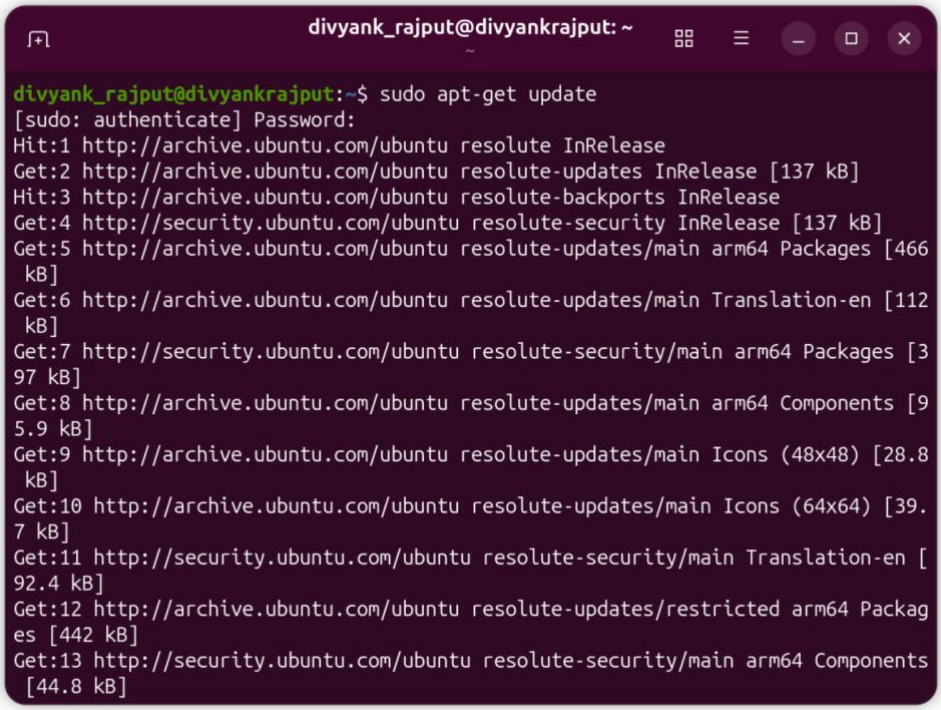


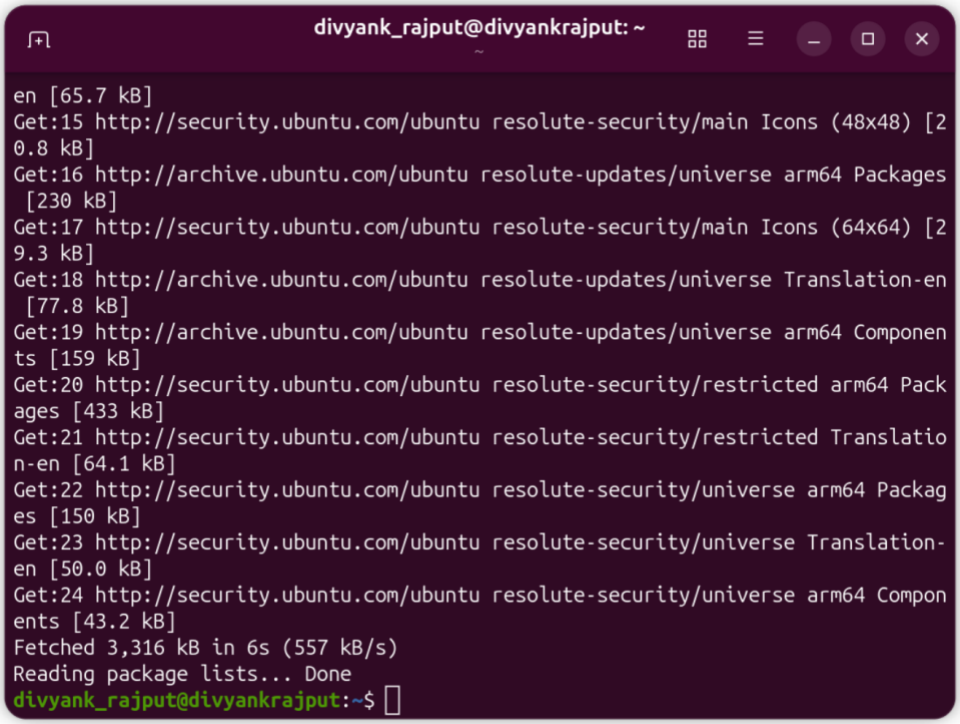
7. How to install, update, upgrade and remove any package in linux (apt-get).

Answers

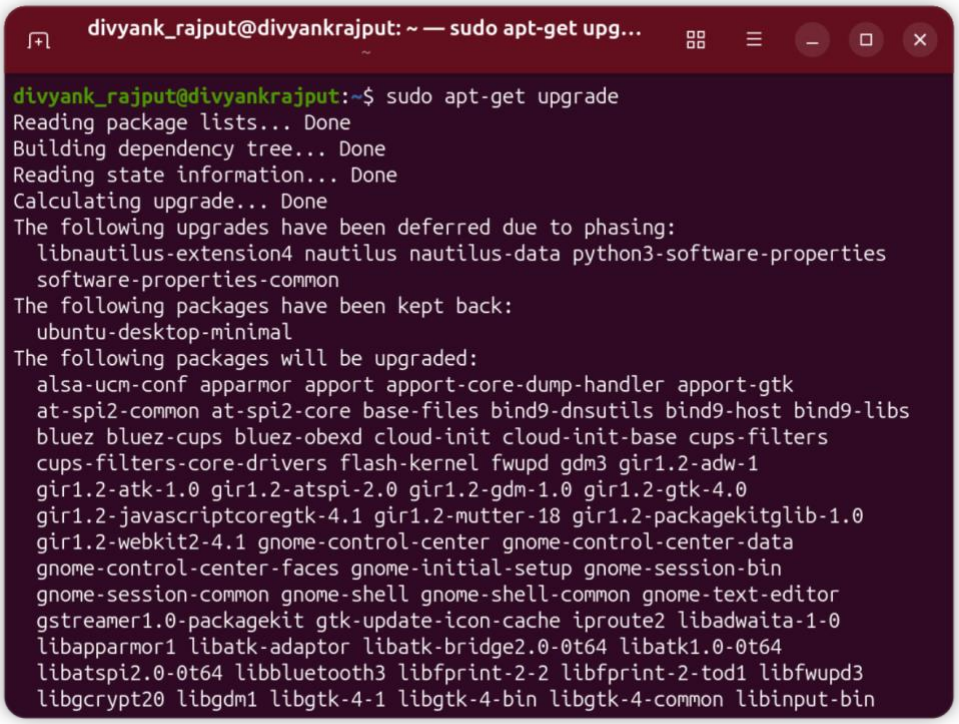
apt-get is a command-line package management tool used in Ubuntu to manage software packages.

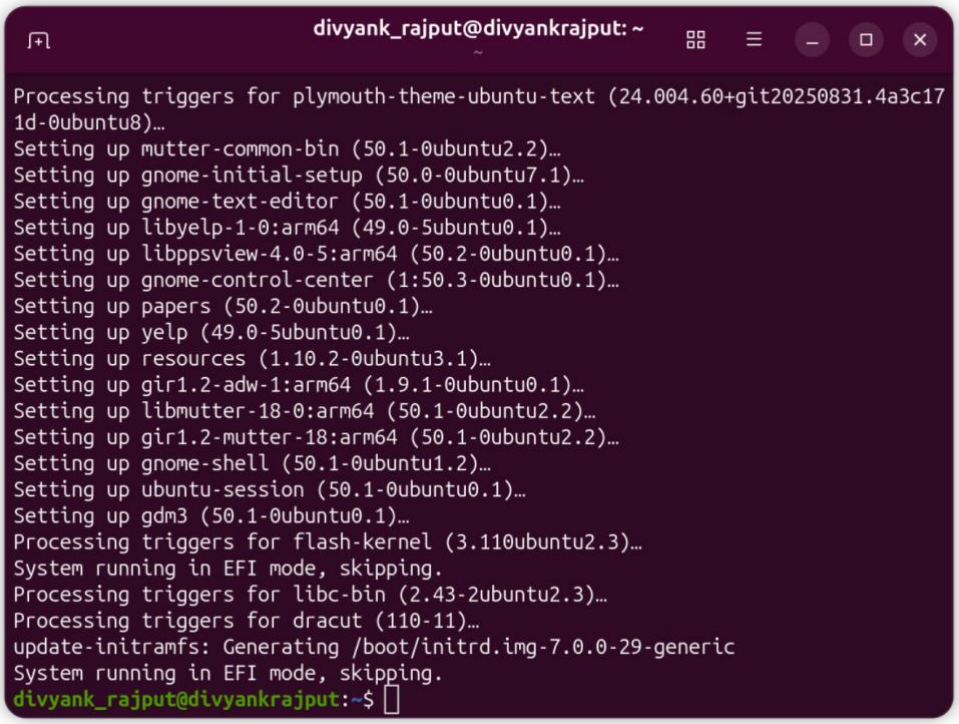
- sudo apt-get update - Updates the list of available packages and their versions



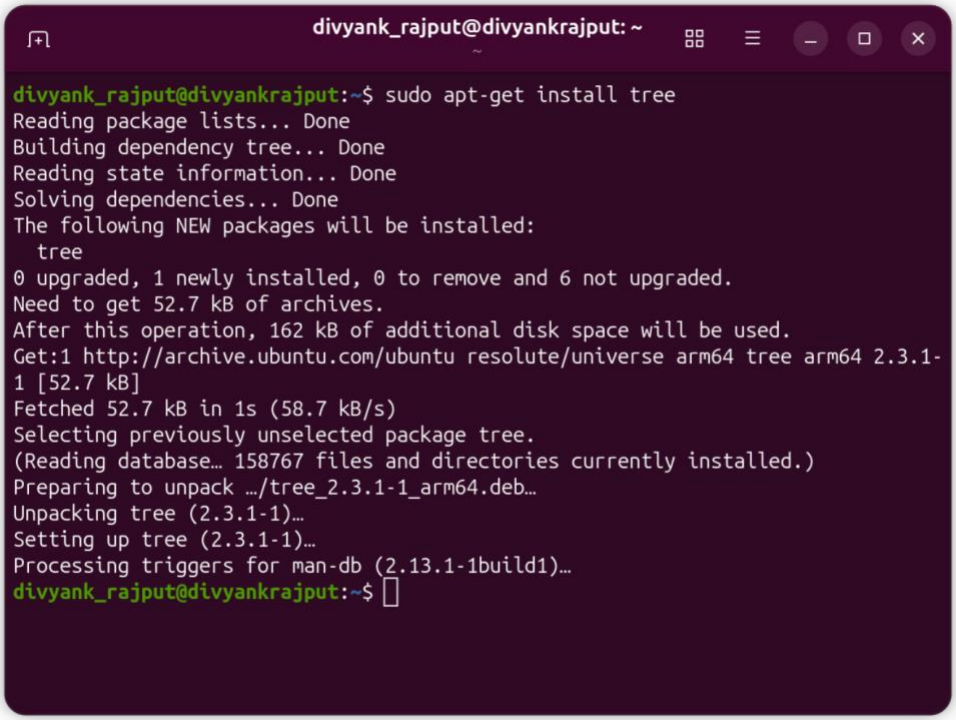


- sudo apt-get upgrade - Upgrades installed packages to their latest available versions.

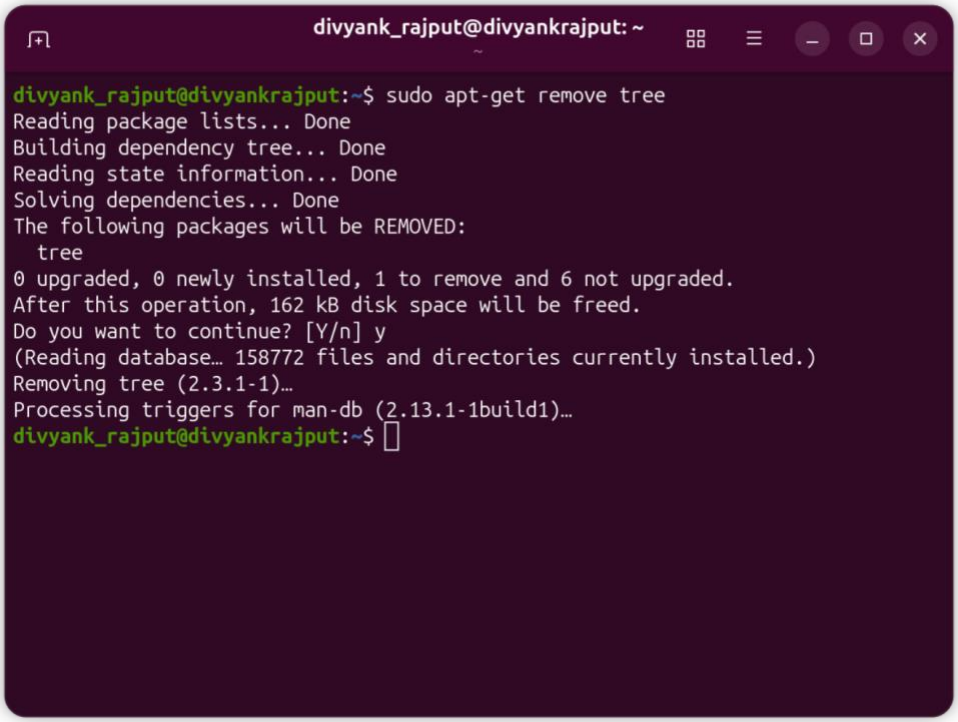




- sudo apt-get install package_name - Installs the specified package.



- sudo apt-get remove package_name - Removes the specified package from the system.

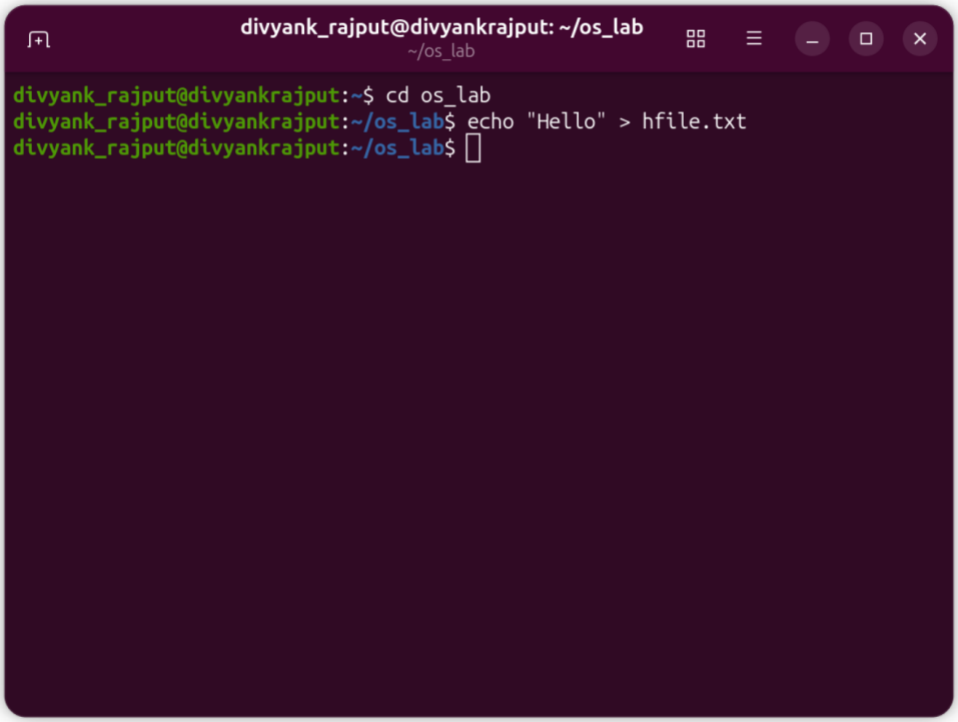


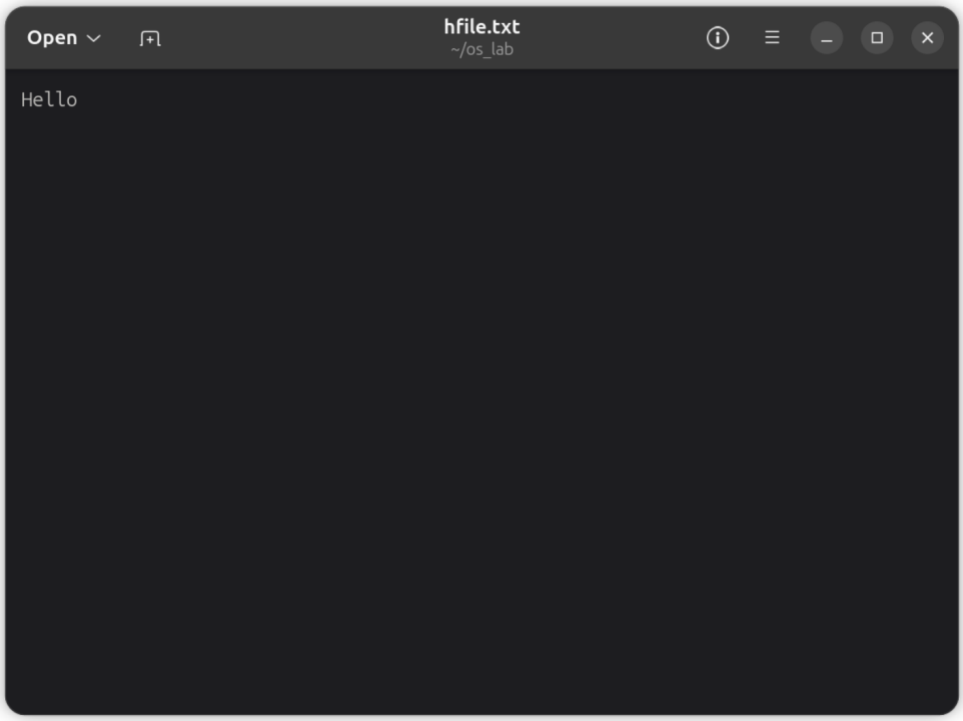
8. >, >> option for directing the output of a command.

Answer

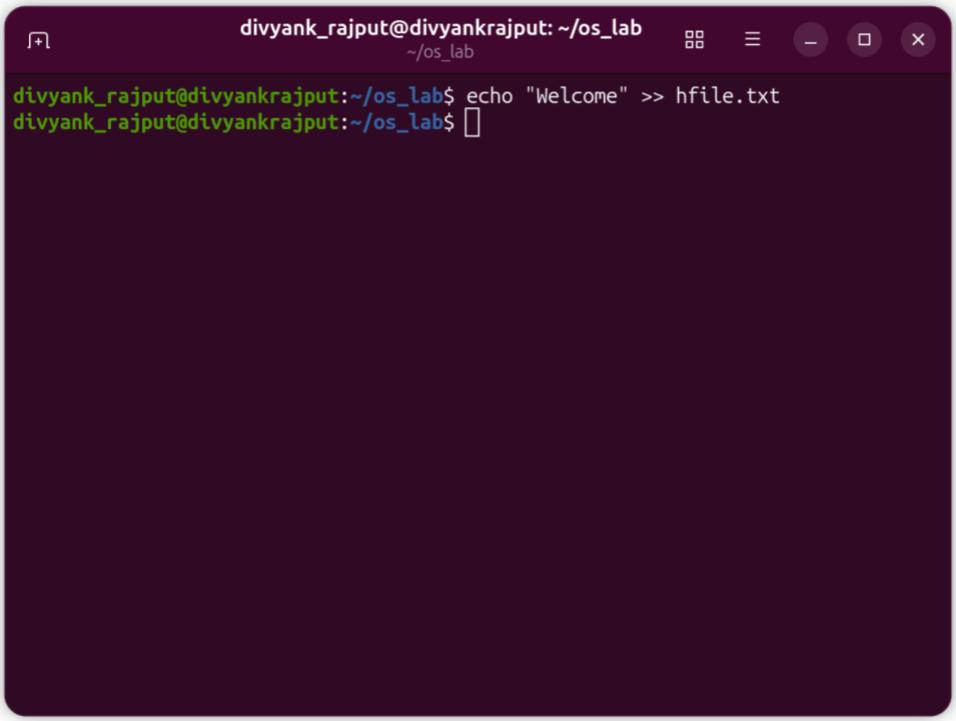
> and >> are used to redirect the output of a command to a file.

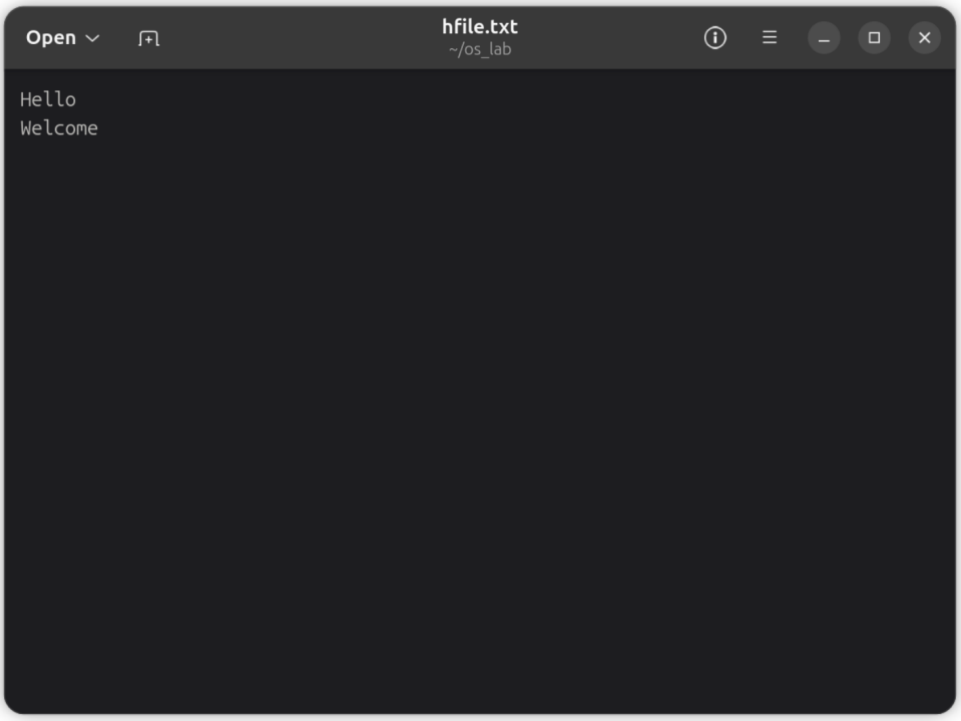
- Redirects the output to a file and overwrites its existing content.





- **>>** - Redirects the output to a file and appends it to the existing content.

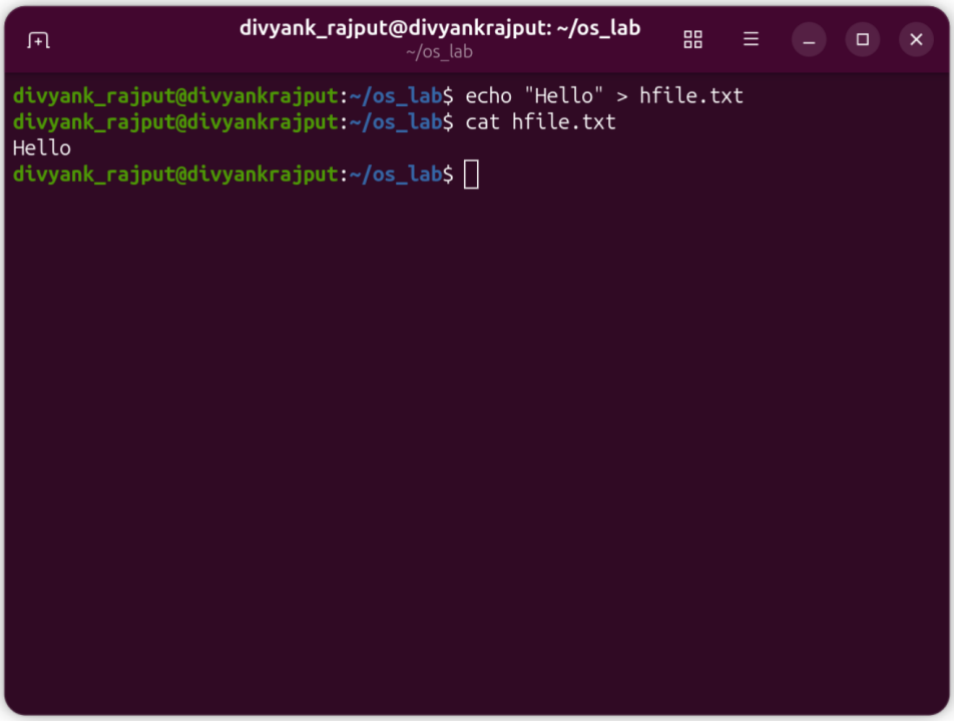

divyank_rajput@divyankrajput: ~/os_lab
~/os_lab
divyank_rajput@divyankrajput:~/os_lab$ echo "Welcome" >> hfile.txt
divyank_rajput@divyankrajput:~/os_lab$

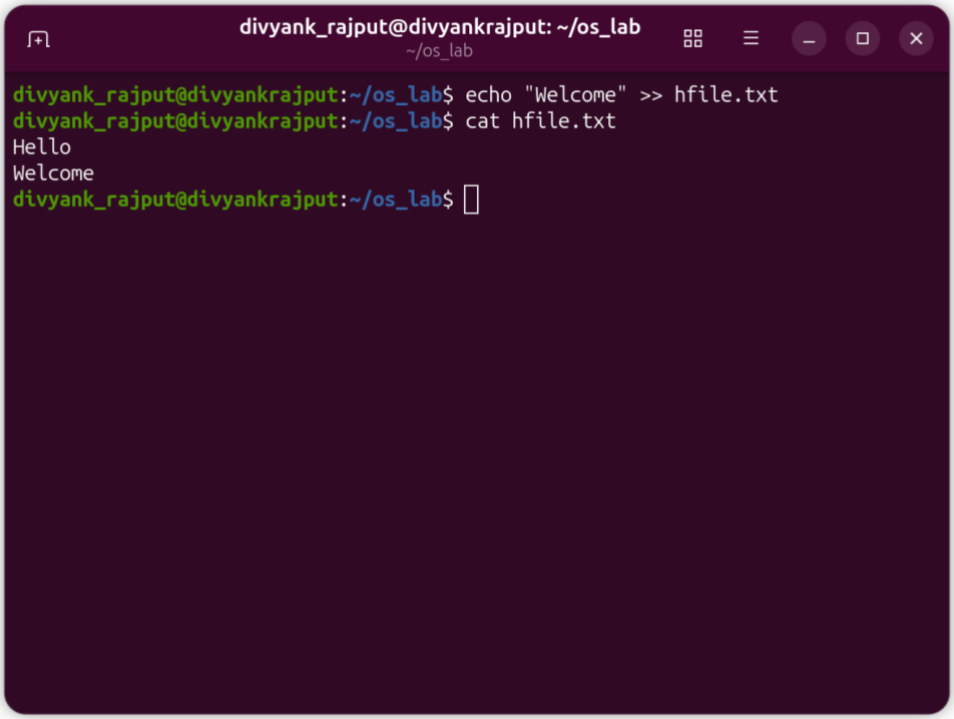

Open
hfile.txt
~/os_lab
Hello
Welcome

9. cat command.

Answer

cat - Used to display the contents of a file on the terminal. It can also be used to create, combine, and display files.

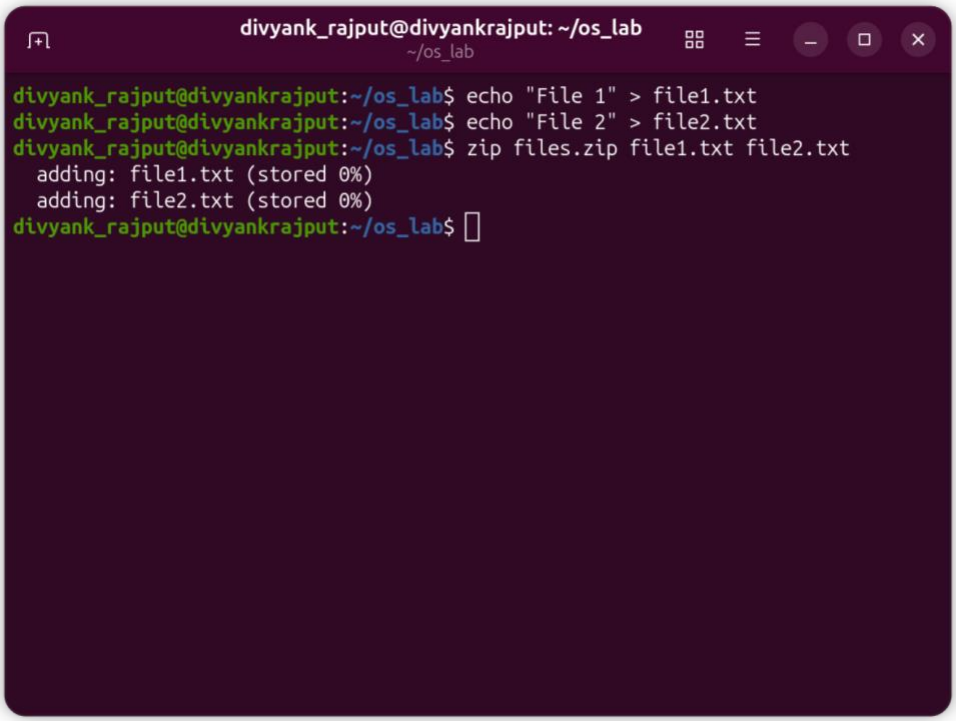




Compressing and archiving files (zip, tar)

Answer

zip - Used to compress and archive files into a .zip file.



tar - Used to combine multiple files and directories into a single archive file.

